

NAME : B. Ganga Jayadev Reddy (192472248)
COURSE NAME : FUNDAMENTALS OF DATA SCIENCE
COURSE CODE : DSA0404

Case Study 1 – Employee Salaries

Data: 4, 5, 5, 6, 7, 8, 8, 9, 30

Q1. Mean

Formula: Mean = $\Sigma x / n$

$$\Sigma x = 4+5+5+6+7+8+8+9+30 = 82$$

$$n = 9$$

$$\text{Mean} = 82/9 = 9.11 \text{ lakhs}$$

Median:

$$\text{Position} = (9+1)/2 = 5\text{th value} = 7 \text{ lakhs.}$$

Mode:

5 occurs twice and 8 occurs twice.

Mode = 5 and 8.

Q2. Range

$$\text{Range} = \text{Maximum} - \text{Minimum} = 30 - 4 = 26 \text{ lakhs.}$$

Q3. Variance & Standard Deviation

$$\text{Mean} = 9.11$$

$(x-\mu)^2$ values:

$$26.11, 16.89, 16.89, 9.67, 4.45, 1.23, 1.23, 0.01, 436.39$$

$$\Sigma(x-\mu)^2 = 512.88$$

$$\text{Variance} = 512.88/9 = 56.99$$

$$\text{Standard Deviation} = \sqrt{56.99} = 7.55$$

$$\text{Q4. Outlier} = 30 \text{ lakhs.}$$

Q5. Median is the best measure because the outlier increases the mean.

Case Study 2 – Website Response Time

Data: 120,125,128,130,132,135,140,145,150,300

Q1. Mean

$$\Sigma x = 1505, n = 10$$

$$\text{Mean} = 1505/10 = 150.5 \text{ ms}$$

Median

$$\text{Median} = (132+135)/2 = 133.5 \text{ ms}$$

$$\text{Q2. Range} = 300 - 120 = 180 \text{ ms.}$$

Q3. Variance & Standard Deviation

$$\text{Mean} = 150.5$$

$$(x - \mu)^2:$$

$$930.25, 650.25, 506.25, 420.25, 342.25, 240.25, 110.25, 30.25, 0.25, 22350.25$$

$$\Sigma(x - \mu)^2 = 25580.5$$

$$\text{Variance} = 25580.5/10 = 2558.05$$

$$\text{Standard Deviation} = \sqrt{2558.05} = 50.58 \text{ ms}$$

Q4. Right-skewed because of the extreme value 300 ms.

Q5. Median is more useful because it is not affected by the outlier.

Case Study 3 – Student Performance

Data: 42,45,48,50,52,52,55,58,60,62,65,95

Q1.

$$Q2 (\text{Median}) = (52+55)/2 = 53.5$$

Lower half: 42,45,48,50,52,52

$$Q1 = (48+50)/2 = 49$$

Upper half: 55,58,60,62,65,95

$$Q3 = (60+62)/2 = 61$$

Q2.

$$IQR = Q3 - Q1 = 61 - 49 = 12$$

Q3.

$$1.5 \times \text{IQR} = 18$$

$$\text{Lower Limit} = 49 - 18 = 31$$

$$\text{Upper Limit} = 61 + 18 = 79$$

$95 > 79$, therefore 95 is an outlier.

Q4.

$$\Sigma x = 684$$

$$\text{Mean} = 684/12 = 57$$

$$\text{Median} = 53.5$$

Q5.

The outlier 95 increases the mean but has little effect on the median.

Case Study 4 – E-Commerce Orders

Data: 18,20,22,22,24,25,26,28,30,35

Q1.

$$\Sigma x = 250$$

$$\text{Mean} = 250/10 = 25$$

$$\text{Median} = (24 + 25)/2 = 24.5$$

$$\text{Mode} = 22$$

Q2.

$$\text{Mean} = 25$$

$(x - 25)^2$ values:

$$49, 25, 9, 9, 1, 0, 1, 9, 25, 100$$

$$\Sigma(x - \mu)^2 = 228$$

$$\text{Variance} = 228/10 = 22.8$$

$$\text{Standard Deviation} = \sqrt{22.8} = 4.77$$

Q3.

$$CV=(4.77/25)\times 100=19.08\%\approx 19.1\%$$

Q4.

Standard deviation shows daily orders vary by about 5 orders from the average.

Q5.

The data is moderately variable.

Case Study 5 – Comparing Two Models

Model A: 2,3,3,4,4,5,5,6

Model B: 1,1,2,4,5,7,8,9

Q1.

$$\text{Model A Mean}=32/8=4$$

$$\text{Model B Mean}=37/8=4.625\approx 4.63$$

Q2.

Model A:

$$\Sigma(x-\mu)^2=12$$

$$\text{Variance}=12/8=1.5$$

Model B:

$$\Sigma(x-\mu)^2=69.875$$

$$\text{Variance}=69.875/8=8.73$$

Q3.

$$\text{Model A SD}=\sqrt{1.5}=1.22$$

$$\text{Model B SD}=\sqrt{8.73}=2.96$$

Q4.

Model A is more consistent because it has the smaller standard deviation.

Q5.

A lower mean error alone is not enough; consistency, variance and overall performance should also be considered.